H

RAFFLES INSTITUTION
2020 YEAR 6 PRELIMINARY EXAMINATION

Higher 3

CHEMISTRY

9813/01

Paper 1

INSERT

30 September 2020

2 hours 30 minutes

INSTRUCTIONS

This insert contains information for Question 1. **Do not write your answers on the insert.**

Information for Question 1

Abstract 1 (*PNAS*, 2008, **105**, 13197 and *Green Chem.*, 2007, **9**, 1273)

Atom Economy

Conventionally, attaining the highest yield and product selectivity were the governing factors of chemical synthesis. Little consideration was given to the usage of multiple reagents in stoichiometric quantities, which often were not incorporated into the target molecule and would result in significant side products. However, in a balanced chemical reaction, a simple addition or cycloaddition incorporates all atoms of the starting materials into the final product. Recognising this fundamental phenomenon, in 1991, Trost presented a set of coherent guiding principles for evaluating the efficiency of specific chemical processes, termed the *atom economy*, which has subsequently been incorporated into the "Twelve Principles of Green Chemistry" and has altered the way many chemists design and plan their syntheses. Atom economy seeks to maximise the incorporation of the starting materials into the final product of any given reaction. The additional corollary is that, if maximum incorporation cannot be achieved, then ideally the quantities of side products should be minute and environmentally innocuous. There is a fundamental difference in the manner in which a reaction yield and the atom economy yield is calculated.

$$\text{reaction yield} = \frac{\text{quantity of product isolated}}{\text{theoretical quantity of product}} \times 100\%$$

$$\text{atom economy} = \frac{\text{molecular weight of desired product}}{\text{molecular weight of all products}} \times 100\%$$

Example: For the reaction $2A + B \longrightarrow C + 3D$, where D is the desired product,

$$\text{atom economy} = \frac{3 \times (M_r \text{ of D})}{M_r \text{ of C} + 3 \times (M_r \text{ of D})} \times 100\%$$

Green Chemistry

Green chemistry efficiently utilises (preferably renewable) raw materials, eliminates waste and avoids the use of toxic and/or hazardous reagents and solvents in the manufacture and application of chemical products. According to this definition, 'raw materials' includes the source of energy. Green chemistry eliminates waste at source, *i.e.*, it is primary pollution prevention rather than waste remediation (end-of-pipe solutions). Prevention is better than cure (the first of the Twelve Principles of Green Chemistry).

Oxidising agents and reducing agents are frequently employed in chemical reactions to transform functional groups into other functional groups. However, the use of oxidising agents and reducing agents contributes to chemical waste that could be harmful to the environment.

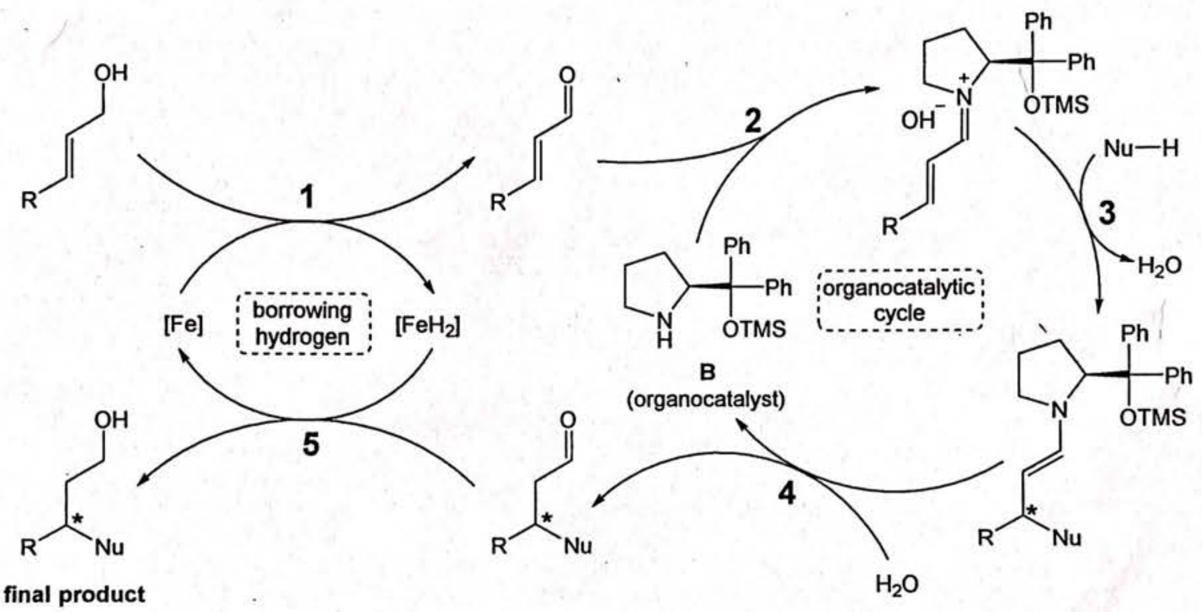
Abstract 2 (*Chem. Rev.*, 2018, **118**, 1410)

The borrowing hydrogen principle, also called hydrogen autotransfer, is a powerful approach which combines transfer hydrogenation (avoiding the direct use of molecular hydrogen) with one or more intermediate reactions to synthesise more complex molecules without the need for tedious separation or isolation processes. The strategy which usually relies on three steps, (i) dehydrogenation, (ii) intermediate reaction, and (iii) hydrogenation, is an excellent and well-recognised process from the synthetic, economic, and environmental point of view. The key to this concept is that the hydrogen from a donor molecule will be stored by a catalytic metal fragment to be released in a final hydrogenation step, hence the reaction name.

Abstract 3 (*Angew. Chem. Int. Ed.*, 2013, **52**, 12883)

In this context, we intended to combine a hydrogen autotransfer process with an organocatalytic cycle in a dual manner. As a result, the reactive carbonyl functionality that is required for an enantioselective aminocatalysed process would be catalytically generated from the alcohol oxidation level (an unactivated functional group), and then transformed *in situ*. The strategy should rely on the use of a metal catalyst that is suitable for borrowing hydrogen processes; if possible, this catalyst should be based on cheap and abundant iron, and it should catalytically and reversibly enable hydrogen transfer without the requirement for a stoichiometric redox reagent. The proposed one-pot relay cascade would be a fully atom economic and waste free transformation of simple allylic alcohols into chiral saturated alcohols, as this process would bypass the three distinct waste producing steps that are otherwise required (oxidation, organocatalytic nucleophilic addition, reduction).

A simplified mechanism of the reaction in Abstract 3 is shown in Fig. 1.1.



R = CH_3, $CH_3CH_2CH_2$, H or Ph
Ph = C_6H_5 (phenyl)
TMS = $(CH_3)_3Si$ (trimethylsilyl)
Nu = nucleophile

Fig. 1.1

Enantioselective Reaction

An enantioselective reaction is a reaction that forms one of the enantiomers preferentially over the other. Enantioselectivity can be achieved with an optically active chiral catalyst.

END OF INSERT

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CANDIDATE NAME		CLASS	2 0 S 0
CENTRE NUMBER	S 3 0 1 6	INDEX NUMBER	

CHEMISTRY

9813/01

Paper 1

30 September 2020

2 hours 30 minutes

Candidates answer on the Question Paper.
Additional Materials: Data Booklet
Insert

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

Section A
Answer **all** questions.

Section B
Answer **two** questions.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

Question	Marks
Section A (answer all)	
1	/ 18
2	/ 11
3	/ 16
4	/ 7
5	/ 8
Section B (answer two)	
6	/ 20
7	/ 20
8	/ 20
Total	/ 100

This document consists of **48** printed pages and **1** insert.

Section A

Answer **all** questions in this section.

1 The information provided in the insert is taken from several published scientific articles. Other published articles may not agree with all of this information.

You should read the whole insert before you start to answer any questions and use the information it contains to answer the questions.

(a) Swern oxidation and Collins oxidation are processes used to oxidise primary and secondary alcohols to aldehydes and ketones respectively. Table 1.1 shows the atom economies for the oxidation of various alcohols into aldehydes or ketones via three different processes.

Table 1.1

	crotyl alcohol (C_4H_7OH)	dodecanol ($C_{12}H_{25}OH$)	cholesterol ($C_{27}H_{45}OH$)
Swern oxidation	18.5%	37.4%	55.5%
Collins oxidation	28.7%	51.4%	68.8%
$K_2Cr_2O_7$ / H_2SO_4	to be calculated	44.4%	62.5%

(i) Potassium dichromate(VI) that is acidified with sulfuric acid can also be used to oxidise primary and secondary alcohols to aldehydes and ketones respectively.

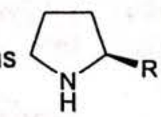
Write the full balanced equation for the oxidation of crotyl alcohol to crotonaldehyde by potassium dichromate(VI) acidified with sulfuric acid. [1]

(ii) Calculate the atom economy for the reaction in **(a)(i)**. [1]

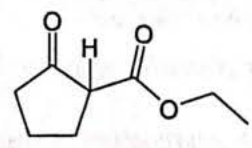
(iii) Comment on the validity of using atom economy to evaluate the efficiency of a chemical process. Support your answer by referencing Table 1.1. [2]

(iv) Suggest how a chemical process might not be green even with 100% atom economy. [1]

(b) (i) Step 2 in Fig. 1.1 was carried out under basic conditions.

Describe the mechanism for step 2. You may represent **B** as  . [2]

Ketoester **A** can be used as the Nu–H in Fig. 1.1.

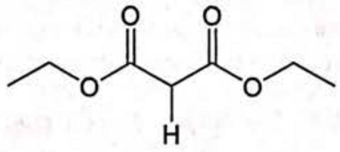


A

The H atom depicted in the given structure of **A** is removed relatively easily by a base to form the nucleophile Nu⁻.

(ii) Draw the final product formed when **A** and crotyl alcohol were reacted as in Fig. 1.1. [1]

(iii) When diethyl malonate was used as Nu–H, the yield was low. It is suggested that this is due to the deprotonation of diethyl malonate being more difficult than that of **A**, so the Nu⁻ is not as easily generated.



diethyl malonate

Explain why the H atom depicted in the structure of diethyl malonate is less easily removed by a base than that in **A**. [1]

(c) In Abstract 3, the intermediate reaction involving the aldehyde was catalysed by organocatalyst **B**.

Fig. 1.2 shows a synthetic route to obtain **B** from L-proline.

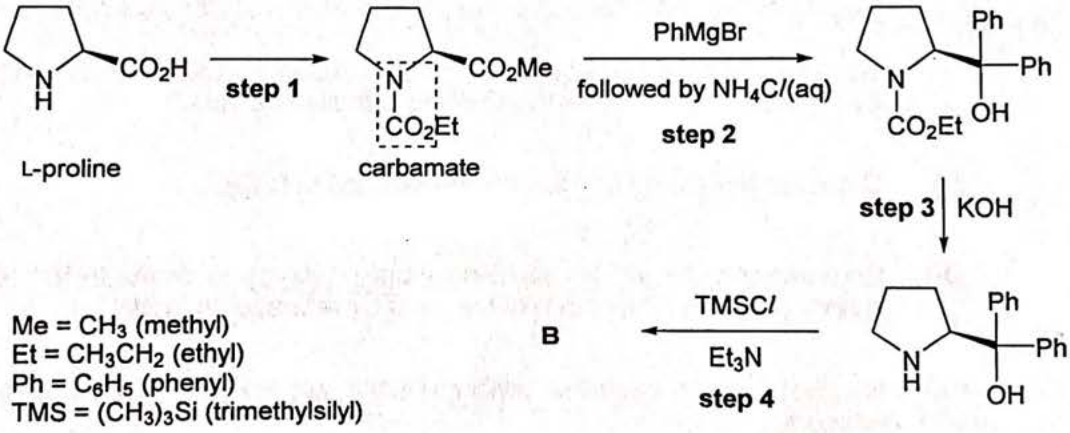


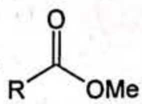
Fig. 1.2

(i) PhMgBr is known as the Grignard reagent. It is a source of the Ph⁻ nucleophile.

If the secondary amine were not converted to a carbamate (–NCO_2–) in step 1 of Fig. 1.2, the yield of step 2 would be low.

Suggest why this is so. [1]

(ii) Describe the mechanism for step 2 in Fig. 1.2.

You may represent the reactant in step 2 as . [2]

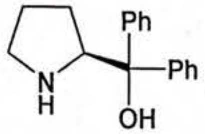
(d) The organocatalyst **B** results in an enantioselective reaction in step 3 of Fig. 1.1. It does this by hindering the approach of the nucleophile from one side due to steric hindrance caused by the two large phenyl rings.

(i) State the stereochemistry (*R* or *S*) of the organocatalyst **B**. [1]

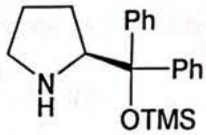
(ii) The product drawn in **(b)(ii)** exhibits enantiomerism. In the scientific paper of Abstract 3, the authors reported obtaining the product with 84% optical purity. The optical rotation observed, $[\alpha]_{obs}$, was −9.3°.

Calculate the optical rotation of the pure major enantiomer, $[\alpha]_{pure}$. [1]

(iii) Poor enantioselectivity was observed when the non-silylated organocatalyst was used in place of **B**.



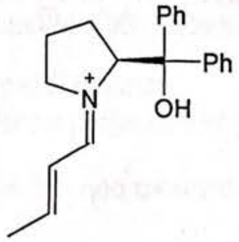
non-silylated organocatalyst



B

By considering the functional groups present on the nucleophile obtained from **A**, suggest why this is so. [1]

(iv) The non-silylated organocatalyst also has an issue of not being able to provide sufficient catalytic turnover. Catalytic turnover refers to the number of times a catalyst can catalyse a reaction.



intermediate formed with
non-silylated organocatalyst

A study revealed that the intermediate shown above will form a side product ($C_{21}H_{23}NO$), which prevents the non-silylated organocatalyst from being regenerated easily. This side product contains two fused five-membered rings. One of the rings contains four carbon atoms and one nitrogen atom, while the other ring contains three carbon atoms, one nitrogen atom and one oxygen atom.

Propose the structure of the side product formed from the intermediate shown above and describe the mechanism for its formation. [2]

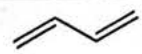
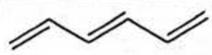
(v) The organocatalyst works in a similar manner to a Lewis acid metal catalyst in activating the aldehyde.

Suggest why using an organocatalyst is greener than using a Lewis acid metal catalyst in synthesis. [1]

[Total: 18]

2 **(a)** Table 2.1 shows two compounds that absorb ultraviolet (UV) radiation.

Table 2.1

compound	structure
buta-1,3-diene	
trans-hexa-1,3,5-triene	

(i) State the type of electronic transition that is responsible for the UV absorptions by the compounds in Table 2.1. [1]

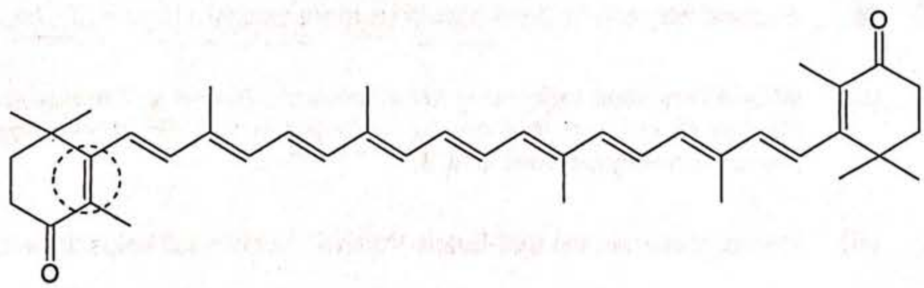
(ii) Draw a molecular orbital diagram for the π system of *trans*-hexa-1,3,5-triene. Label the HOMO and LUMO clearly. [3]

During a process known as photoisomerism, *trans*-hexa-1,3,5-triene can transform to *cis*-hexa-1,3,5-triene reversibly, when either isomer absorbs a photon that matches the energy gap for the electronic transition stated in **(a)(i)**.

(iii) Using your molecular orbital diagram drawn in **(a)(ii)**, calculate the π bond order before and after a photon is absorbed. [1]

(iv) Hence, explain why *trans*-hexa-1,3,5-triene can isomerise to *cis*-hexa-1,3,5-triene when exposed to light of a certain wavelength. [2]

(b) Carrots are a common source of β-carotene.



β-carotene

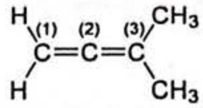
(i) Assign the stereochemistry (*E* or *Z*) for the double bond that is circled. [1]

(ii) By considering the structure of β-carotene, explain why it is coloured. [1]

(iii) Explain why carrots are decolourised after they are soaked in aqueous bromine. [2]

[Total: 11]

3 Allenes are organic compounds in which one carbon atom forms double bonds with each of its two adjacent carbon atoms. 3-methylbuta-1,2-diene is an example of an allene.



3-methylbuta-1,2-diene

(a) The IR spectrum of 3-methylbuta-1,2-diene shows strong absorbance at around 1960 cm^{-1}.

(i) Explain the origin of IR absorptions of simple molecules. [2]

(ii) State the hybridisation of C(1), C(2) and C(3). [1]

(iii) The IR absorption frequency of an alkene C=C bond differs quite significantly from that of an allene.

Using relevant information from the *Data Booklet*, deduce the relative strength of an allene C=C bond compared to an alkene C=C bond.

Explain the relative strength of the bonds using your answer to **(a)(ii)**. [2]

(b) 3-methylbutyne, an isomer of 3-methylbuta-1,2-diene, shows three signals in its ^{1}H NMR spectrum. Explain why the singlet signal resides at a low chemical shift δ of around 1.8 ppm. [2]

(c) Penta-2,3-diene is an isomer of 3-methylbuta-1,2-diene. It exhibits enantiomerism despite the absence of a chiral centre.

(i) Draw the structures of both enantiomers, illustrating the stereochemistry clearly with wedge and hash bonds. [1]

(ii) The stereochemistry (*R* or *S*) of an allene can be determined by using the Newman projection of the allene, looking through the C=C=C bond. The priorities of substituents on both ends of the allene are assigned according to Cahn-Ingold-Prelog rules. This is illustrated in Fig. 3.1.

Fig. 3.1

Draw the Newman projections of both enantiomers from **(c)(i)** and assign the stereochemistry (*R* or *S*) to each of them. [2]

(d) In the electrophilic addition reaction between buta-1,3-diene and hydrogen chloride, the major product formed is 3-chlorobut-1-ene.

(i) Draw the structure of the carbocation **C3** that leads to the major product and explain how **C3** is stabilised. [2]

Buta-1,2-diene also reacts with hydrogen chloride via the same mechanism. The major product formed is 2-chlorobut-2-ene instead of 1-chlorobut-2-ene, as shown in Fig. 3.2.

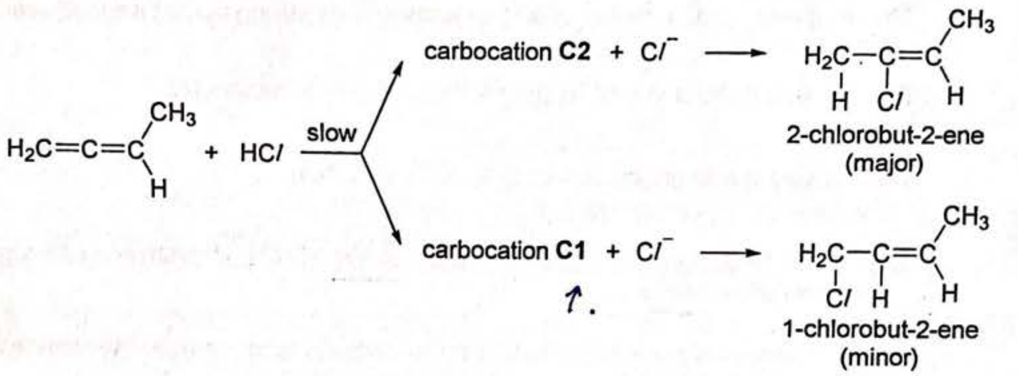


Fig. 3.2

(ii) By considering the orientation of the π electron clouds that make up the C=C=C in buta-1,2-diene, draw the structure of **C1**, showing the orientation of the p orbitals on all three carbon atoms clearly.

Hence, comment on the expected stability of **C1** relative to that of **C3**. [2]

(iii) The energy profile diagram for the slow step of the above electrophilic addition reaction is shown in Fig. 3.3.

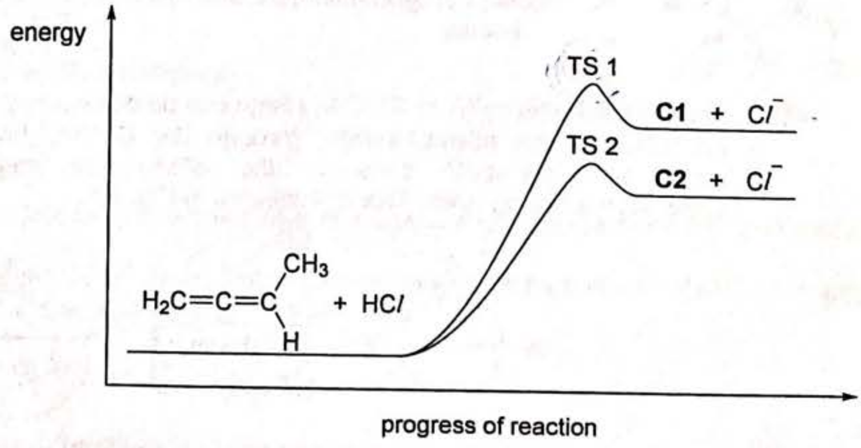


Fig. 3.3

State Hammond's Postulate and hence explain why TS 1 is higher in energy than TS 2. [2]

[Total: 16]

4 Electrophilic substitution reactions on benzene rings have been widely studied, especially the regioselectivity of the products formed.

In the reaction between methylbenzene and Br^+ to form the carbocation, the activation energy, E_a, and the enthalpy change of reaction, ΔH_r, depend on the position (2-, 3- or 4-) to which Br adds to form the carbocation.

Table 4.1 shows the results when Br^+ adds to the various positions of methylbenzene to form the respective carbocations.

Table 4.1

position	ΔH_d / kJ mol^{-1}	E_a / kJ mol^{-1}
2-	−14.7	+84.3
3-	+1.75	+95.9
4-	−19.4	to be calculated

Note:
ΔH_d does not give the enthalpy change between reactants (methylbenzene and Br^+) and product (carbocation), which is denoted by ΔH_r.

$$\Delta H_d = H_x - H_1$$

H_1 is the **absolute enthalpy** of the carbocation with Br at the 1-position; and
H_x is the **absolute enthalpy** of the carbocation with Br at the x-position (x = 2, 3 or 4).

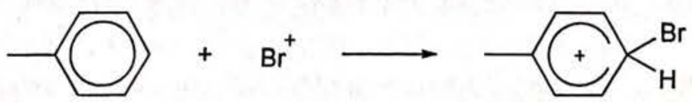
The Bell-Evans-Polanyi Principle describes a mathematical relationship between the activation energies of a series of related reactions and their enthalpy changes.

(a) **(i)** Suggest why ΔH_d for the 4-position is more exothermic than ΔH_d for the 2-position. [1]

(ii) Write an equation expressing ΔH_r in terms of ΔH_d and ΔH_1, where ΔH_1 is the enthalpy change of reaction to form the carbocation with Br at the 1-position from the reactants methylbenzene and Br^+. [1]

(iii) Hence, show that the Bell-Evans-Polanyi Principle still holds between ΔH_d and E_a. [1]

(b) Calculate a value for the activation energy of the following reaction, using the data in Table 4.1.



[3]

(c) The Bell-Evans-Polanyi Principle does not hold for the nitration of methylbenzene.

Explain the significance of this result. [1]

[Total: 7]

5 Derivatives of compound **E** are used to treat various psychiatric disorders such as schizophrenia. Compound **E** contains elements C, H and O only. Its mass spectrum and ^{1}H NMR spectrum are shown in Fig. 5.1 and Fig. 5.2 respectively. The molecular ion peak in its mass spectrum is at m/z 148, and the ratio of the M and M+1 peak heights is 55 : 6.

The peak areas in the ^{1}H NMR spectrum are shown in brackets beneath each peak.

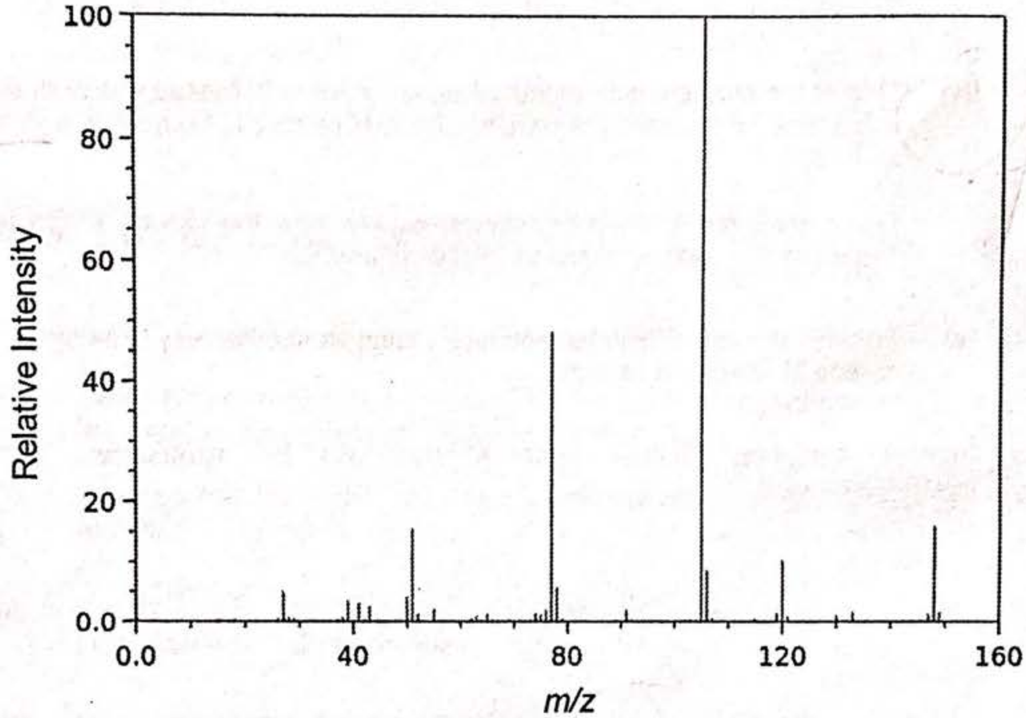


Fig. 5.1

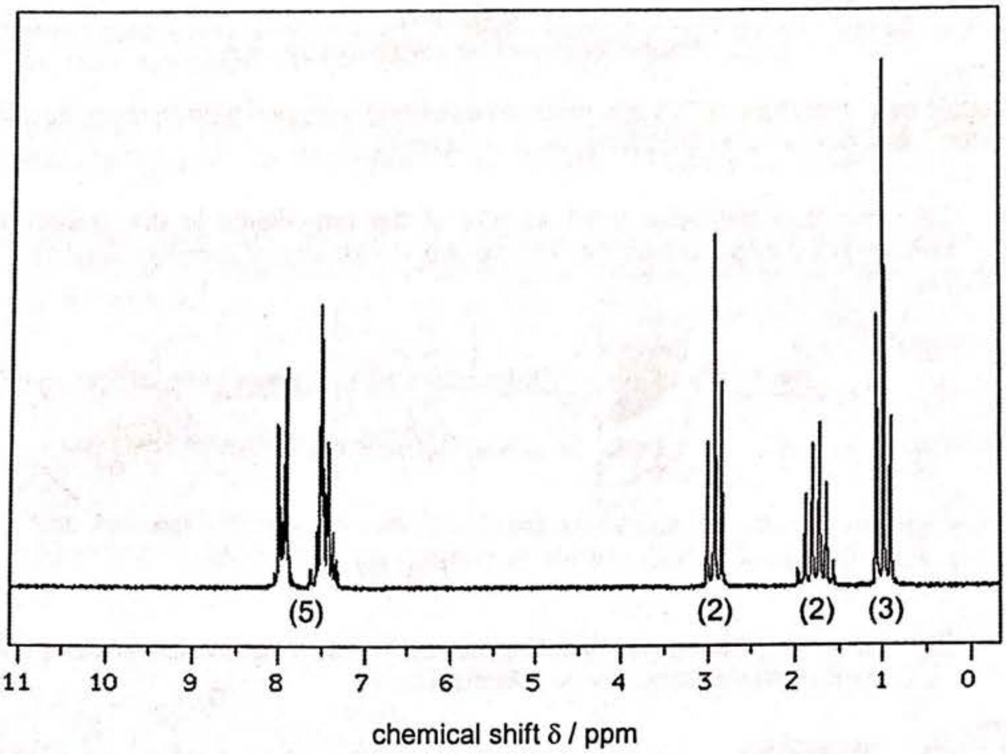


Fig. 5.2

(a) Deduce the molecular formula of **E**, showing your reasoning. [2]

(b) Deduce the structure of **E** using information from the ^{1}H NMR spectrum. [3]

(c) **(i)** Suggest the structure of the fragment at *m/z* 105. [1]

(ii) Hence, explain why the relative intensity at *m/z* 105 is high. [1]

(d) Suggest the structure of the fragment at *m/z* 120. [1]

[Total: 8]

Section B

Answer **two** questions from this section.

7 Linalool pyrophosphate (LPP) is a naturally occurring terpene found in many flowers and spice plants. It is a precursor to the synthesis of terpenoids.

(a) Camphor is a terpenoid used as one of the ingredients in the manufacture of pest insecticides and preservatives. The synthesis pathway of camphor from LPP is shown in Fig. 7.1.

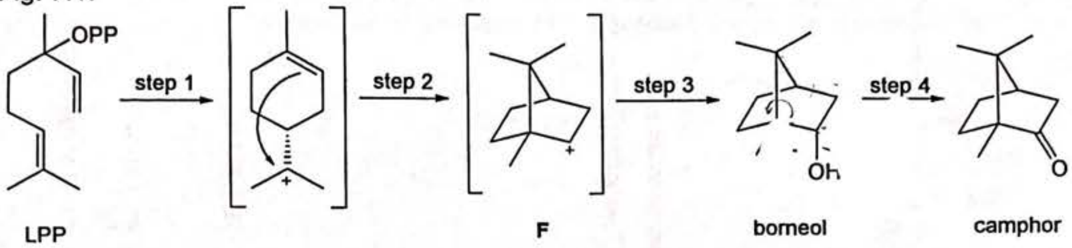


Fig. 7.1

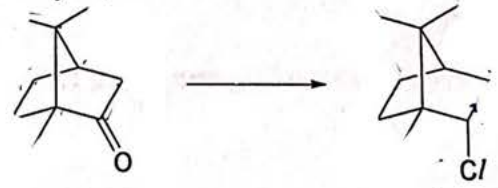
(i) Draw another intermediate that can be formed in step 2. Explain why it is **not** formed preferentially compared to intermediate **F**. [2]

(ii) On Fig. 7.1, assign the stereochemistry (*R* or *S*) of all chiral centres of borneol. [1]

(iii) State the number of stereoisomers possible for borneol, including borneol itself. [1]

(iv) Compound **J** is a precursor to an organometallic catalyst, and it can be formed from camphor.

Suggest a synthetic route to convert camphor to **J**. Your answer should clearly show the stereochemistry of any intermediates formed.



[2]

(v) Briefly describe the differences between the molecular ion peak of camphor and that of **J** in their respective mass spectra. [2]

(b) **J** can undergo elimination when heated with sodium hydroxide to form an alkene, **M**. The steps involved in this reaction are shown in Fig. 7.2.

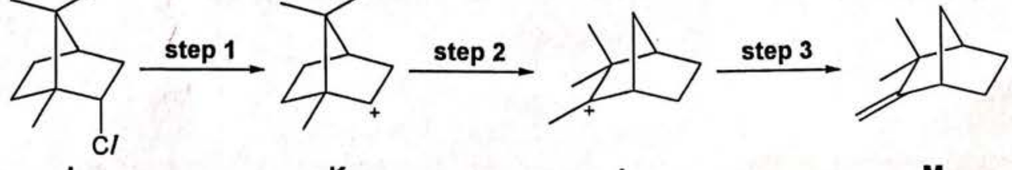


Fig. 7.2

(i) Step 2 involves a rearrangement reaction. Draw curly arrows on **K** in Fig. 7.2 to illustrate this rearrangement. [1]

(ii) Explain why **M** is the only alkene product formed in step 3. [1]

(iii) Draw an energy profile diagram for the formation of **M** from **J**.

Your diagram should clearly show the relative energies of **J**, **K**, and **L**. [2]

(iv) One of the possible side products formed when **J** is heated with sodium hydroxide is borneol. To increase the yield of alkene **M** relative to borneol, a high temperature is used.

Using concepts of thermodynamics, explain how the use of a high temperature increases the yield of alkene **M** relative to borneol. [2]

(v) Besides the use of high temperatures, suggest another way to increase the yield of alkene **M** relative to borneol. [1]

(c) Besides camphor, another terpenoid that can be synthesized from LPP is isodihydrocarveol.

camphor isodihydrocarveol

(i) Describe two differences between the ^{1}H NMR spectra of camphor and isodihydrocarveol. [2]

(ii) One signal in the ^{1}H NMR spectrum of isodihydrocarveol disappears when D_2O is added to its NMR sample.

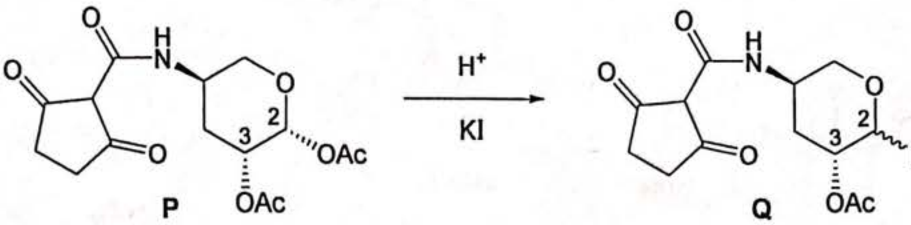
Explain why this signal disappears and include an equation in your explanation. [1]

(iii) Draw the most stable chair conformation of isodihydrocarveol. [1]

(iv) Hence, explain why isodihydrocarveol undergoes E2 elimination at a slow rate. [1]

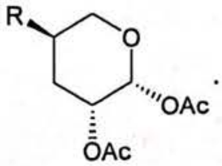
[Total: 20]

8 Fig. 8.1 shows a substitution reaction with compound **P** as the reactant.



where –OAc = –O–$COCH_3$

Fig. 8.1

In all the answers to **8(a)**, you may represent **P** as 

(a) **(i)** Two stereoisomers of **Q** were formed in a 9:1 ratio. The predominant mechanism for the reaction was determined to be S_N2 and not S_N1.

Describe this S_N2 mechanism and predict the structure of the major stereoisomer formed. [3]

(ii) By considering the structure of **P**, suggest a reason why the reaction can also proceed via an S_N1 mechanism. [1]

(iii) Draw the chair conformer of **P** that is required for the S_N2 mechanism described in **(a)(i)** and comment on its expected stability. [2]

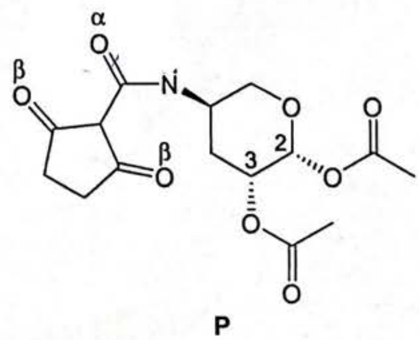
(iv) By considering the orientation of the non-bonding orbital on the O atom of the ring and the anti-bonding orbital of the C2–OAc bond, suggest why the chair conformer of **P** in **(a)(iii)** is more stable than expected. [1]

(v) Suggest reasons why both S_N1 and S_N2 do not take place at C3 of **P**, when it adopts the conformation drawn in **(a)(iii)**. [2]

(b) Table 8.1 shows some of the IR absorptions of **P**.

Table 8.1

wavenumber / cm^{-1}	strength	appearance
3450	w	very broad
3300	m	very broad
1750	s	sharp
1691	s	–
1634	s	–



P

When oxygen α was labelled with ^{18}O, the IR absorption at 3300 cm^{-1} shifted to 3130 cm^{-1}, while the IR absorption at 1634 cm^{-1} shifted to 1607 cm^{-1}.

When oxygen β was labelled with ^{18}O, the IR absorptions at 3300 cm^{-1} and 3450 cm^{-1} shifted slightly, while the IR absorption at 1691 cm^{-1} shifted to 1670 cm^{-1}.

(i) **P** equilibrates with its tautomer. Use the structure of **P** and the information involving ^{18}O labelling to identify the functional groups responsible for all of the IR absorptions in Table 8.1. [2]

(ii) Suggest the structure of the most stable tautomer of **P**.

Use the information given to justify your structure. [3]

Compound **R** is an aromatic compound. The Br atom could be substituted with the $-NH_2$ group via the mechanism shown in Fig. 8.2, instead of via the typical nucleophilic aryl substitution mechanism.

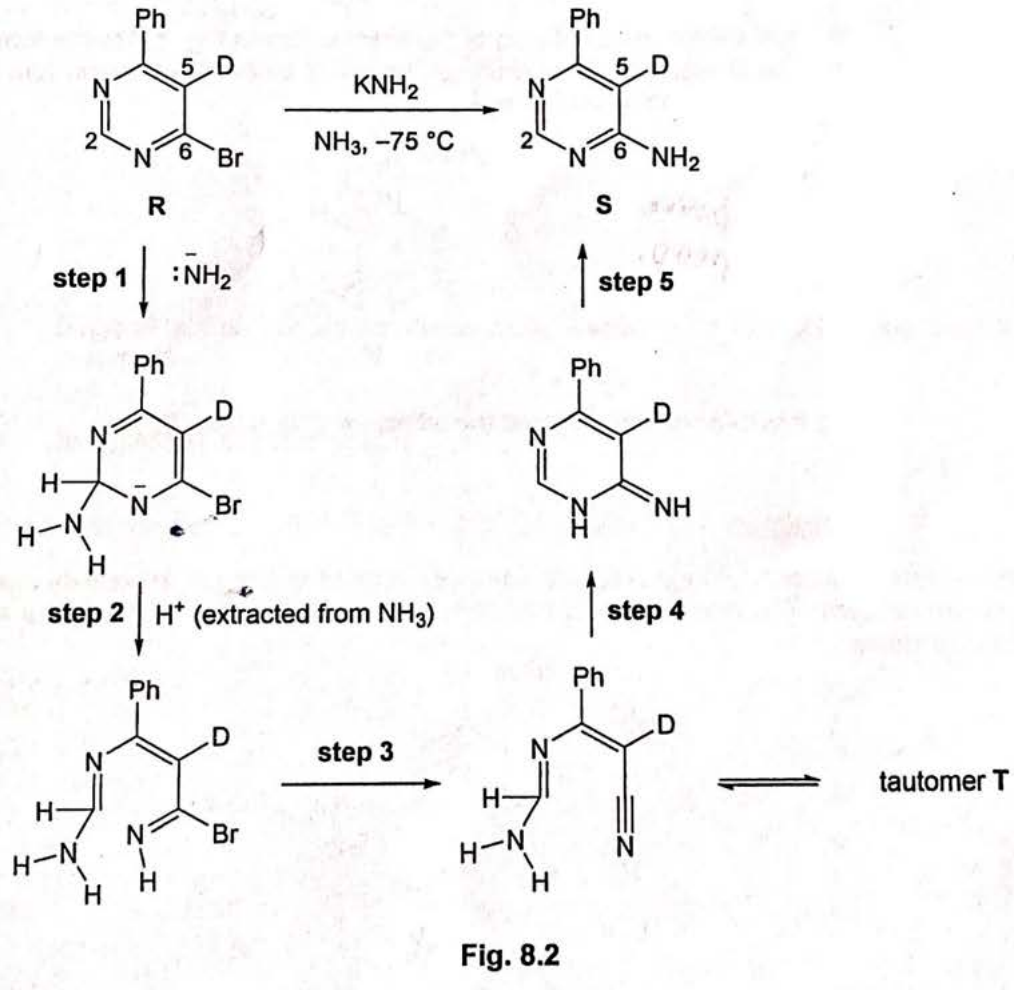


Fig. 8.2

(c) Table 8.2 gives some data from the UV spectra for compounds **R** and **S**.

Table 8.2

	compound **R**	compound **S**
ε_{280} / L mol^{-1} cm^{-1}	12750	6500

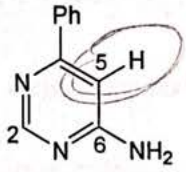
The absorbance, at λ = 280 nm, of the reaction mixture in a cell of length 1.0 cm decreases from 1.50 at the start of the reaction to 1.09 at the end of the reaction.

Calculate the percentage yield of the reaction. [1]

(d) **(i)** Suggest why nucleophilic attack in step 1 takes place on C2 rather than on C6. [1]

(ii) Describe the mechanisms of steps 1 and 2. Use curly arrows to illustrate the movement of electrons. [3]

(iii) A piece of evidence that supports the mechanism in Fig. 8.2 is the formation of **S'**, where the D atom on C5 is replaced by the H atom. The D atom can leave as an exchangeable atom in tautomer **T**.

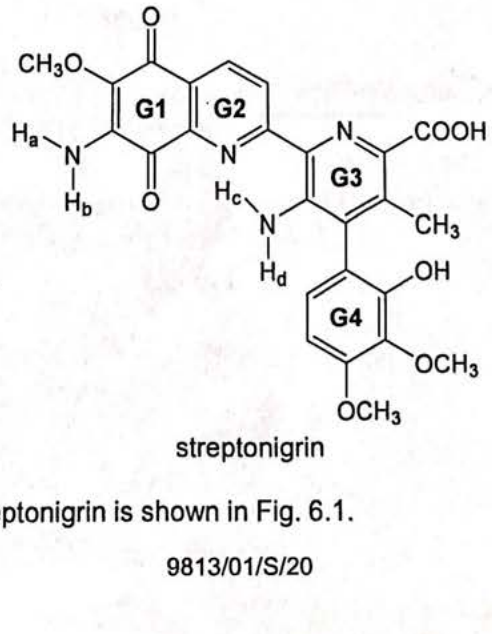


S'

Using this information, suggest the structure of tautomer **T**. [1]

[Total: 20]

6 Streptonigrin is an anti-tumour antibiotic which exhibits broad spectrum activity against a range of human cancers. The crystal structure of streptonigrin shows that rings **G1**, **G2** and **G3** lie on the same plane.



streptonigrin

Part of the synthesis of streptonigrin is shown in Fig. 6.1.

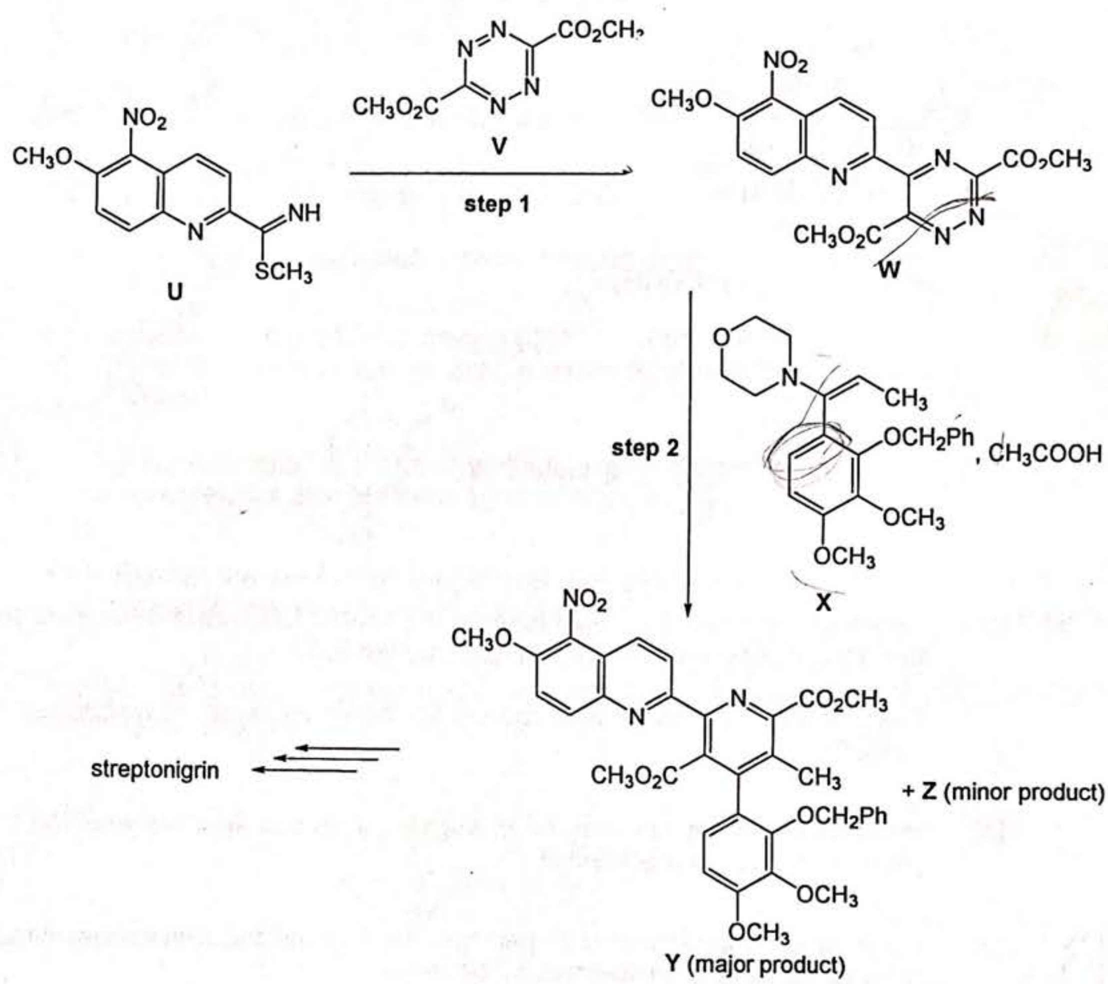


Fig. 6.1

(a) A reaction happens when the HOMO of one reactant interacts with the LUMO of another reactant.

The reaction between **U** and **V** in step 1 of Fig. 6.1 involves a cycloaddition reaction that is similar to that shown in Fig. 6.2.

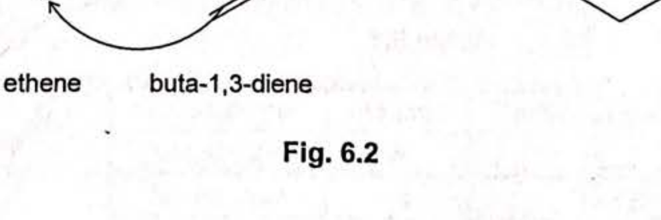


Fig. 6.2

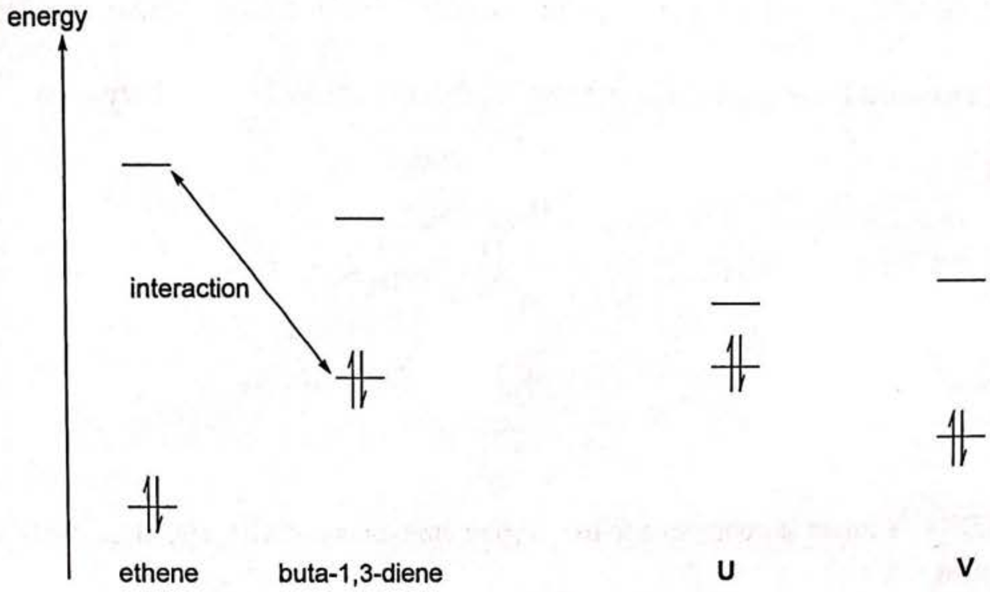


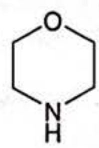
Fig. 6.3

The HOMOs and LUMOs of the reactants for both reactions are shown in Fig. 6.3.

(i) On Fig. 6.3, show clearly which molecular orbitals are interacting for the reaction between **U** and **V**. [1]

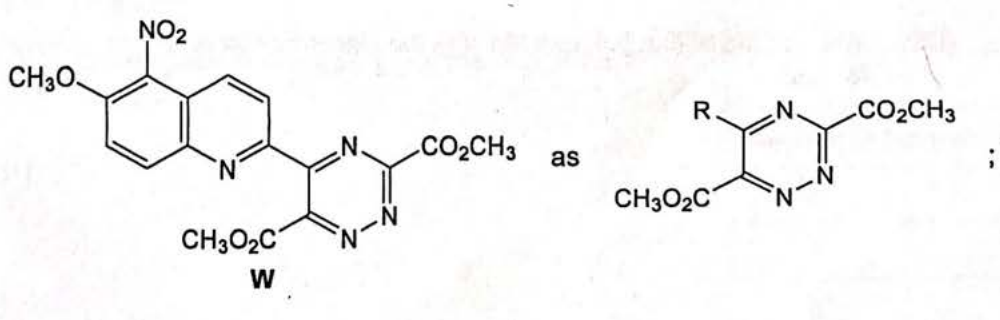
(ii) Hence, deduce whether the reaction between **U** and **V** takes place more readily than that between ethene and buta-1,3-diene. [2]

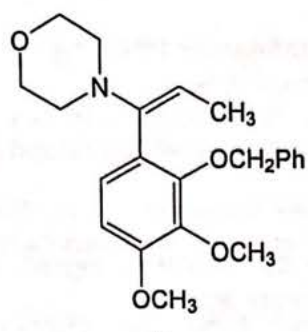
(b) In step 2 of Fig. 6.1, **W** and **X** undergo a similar reaction to step 1. Nitrogen gas and morpholine are formed as by-products of the reaction.



morpholine

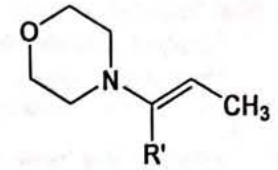
In the answers to **6(b)(i)** and **6(b)(ii)**, you may represent



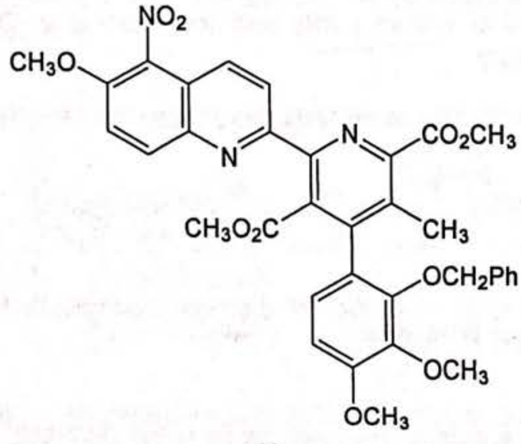


X

as



; and



Y

as

.

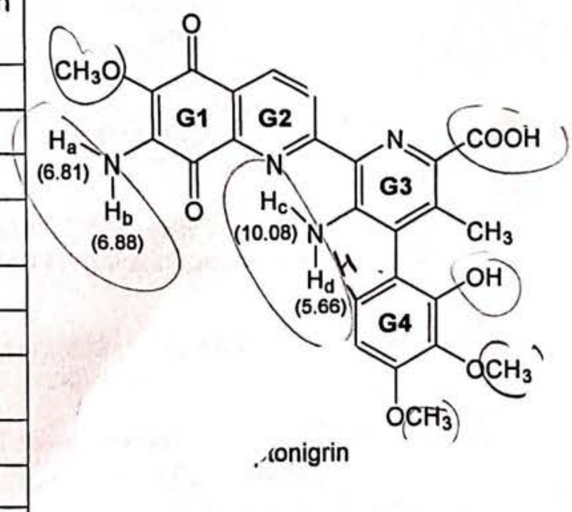
(i) Suggest a mechanism for the reaction between **W** and **X** in step 2 of Fig. 6.1 to form **Y**. [3]

(ii) Suggest the structure of **Z**. [1]

(c) Table 6.1 shows the signals in the ^{1}H NMR spectrum of streptonigrin.

Table 6.1

signal label	chemical shift δ / ppm	splitting pattern	integration value
S1	2.18	singlet	3H
S2	3.16 – 3.20	–	9H
S3	5.66	singlet	1H
S4	6.70 – 6.73	pair of doublets	2H
S5	6.81	singlet	1H
S6	6.88	singlet	1H
S7	8.36 – 8.92	pair of doublets	2H
S8	9.01	singlet	1H
S9	10.08	singlet	1H
S10	12.32	singlet	1H



..tonigrin

(i) H_a, H_b, H_c and H_d are responsible for the signals at chemical shift δ 6.81, 6.88, 10.08 and 5.66 ppm respectively.

Explain why the difference in chemical shift δ between H_c and H_d is so much greater than that between H_a and H_b. [2]

(ii) Identify the protons responsible for the remaining signals and account for the splitting patterns, if any, observed in these signals. [4]

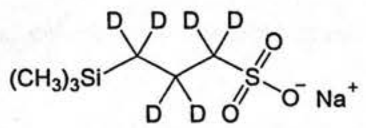
(d) To determine the purity of a sample of synthesised streptonigrin using [1]H NMR, the sample is mixed with an internal standard of known purity and the mixture is dissolved in a deuterated solvent.

A quantitative value of the sample's purity can be calculated using the following equation.

$$\%P_{sample} = \frac{S_{sample}}{S_{std}} \times \frac{N_{std}}{N_{sample}} \times \frac{m_{std}}{m_{sample}} \times \frac{M_{sample}}{M_{std}} \times \%P_{std}$$

where S = integrated area of peak; N = number of protons responsible for the peak; m = mass used; M = molecular mass; P = purity.

A student determined the purity of a sample of streptonigrin using [1]H NMR. The internal standard used was DSS.



DSS

Table 6.2

	DSS	streptonigrin
chemical shift δ of peak used / ppm	0.10	2.18
mass used / mg	2.00	5.00
molar mass / g mol^{-1}	224.3	506.5
percentage purity	99.95%	to be calculated

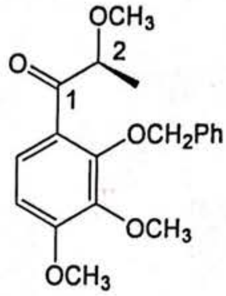
(i) Suggest why DSS is a suitable internal standard for the determination of purity of streptonigrin using [1]H NMR. [1]

(ii) The ratio of the integrated area of the peak at δ 2.18 ppm to that of the reference peak of DSS was found to be 0.280.

Using the information in Table 6.2, calculate the percentage purity of the sample of streptonigrin. [1]

(e) State the type of stereoisomerism that streptonigrin can exhibit. Explain your answer. [2]

(f) Compound **H** undergoes reduction with $ZnBH_4$ to form 99% of one stereoisomer.



H

Zn^{2+} was found to coordinate to the oxygen atom of the $-OCH_3$ and of the C=O of **H**.

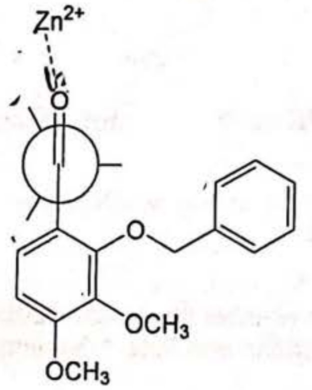


Fig. 6.4

You may assume that $ZnBH_4$ is a source of H^- ions.

(i) On Fig. 6.4, complete the Newman projection of **H** by filling in the groups on C2.

Use a dotted line to clearly show the coordination between the oxygen atom of the $-OCH_3$ group and Zn^{2+}. [1]

(ii) Draw the structure of the major stereoisomer formed from the reduction with $ZnBH_4$, showing the stereochemistry clearly. [1]

(iii) With the aid of Fig. 6.4, explain why the stereoisomer drawn in **(f)(ii)** is preferentially formed. [1]

[Total: 20]